

A Crash Response System using LoRa-based V2X Communications

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Abstract—Emergency services response time is a critical factor for road traffic injury care, and the current scenario is that most accidents must be reported by a bystander or by those involved, which usually takes several minutes. This work proposes a crash response system using LoRa-based vehicular communications, aiming to reduce response times for road accidents, save lives, and take advantage of the license-free radio frequencies used by LoRa. The system uses broadcast beacons to build multi-hop routes from vehicle nodes to emergency services. The beacon message contains important information, such as the type of accident, latitude, longitude, the estimated number of people, and the last known speed. We have evaluated our proposal in the ns-3 network simulator by simulating an urban environment in a 10km wide area and varying the node density. Results have shown that the network can transmit a beacon across several kilometers with minimal latency. The amount of nodes in the network is a critical factor, with greater node density network instances displaying a larger Packet Delivery Ratio (PDR) and Network Coverage but also a slightly longer latency. As a complementary contribution, all the code produced in this research is publicly available as Open Source Software.

Index Terms—Emergency Response System, LoRa, V2X.

I. INTRODUCTION

In June of 2022, the World Health Organization (WHO) released an article claiming that “road traffic injuries are the leading cause of death for children and young adults aged 5-29 years” [1], and that more than one million people die each year as a result of road traffic crashes. Also according to the same organization, “emergency care for injury is at the core of the post-crash response” and “effective care of the injured requires a series of time-sensitive actions” [2].

A road accident is usually followed by a notification to the emergency services done by a bystander, dispatching of the medical team, arrival and care at the scene, before transfer to a medical facility [3]. However, the European Transport Safety Council (ETSC) writes that “there is usually a delay until the incident has been recognised by those involved or bystanders. This delay is followed by the time it takes to alert the relevant emergency services”.

Answering these needs, in April 2018 the automatic dialling system *eCall* was introduced in the EU, being mandatory for

all new cars. It is expected that *eCall* “will cut emergency service response time by 50% in the countryside and 60% in built-up areas and the quicker response will save hundreds of lives in the EU every year” [4].

Central to the success of any technology oriented to reduce the delay of the notifications to road emergency systems, such as *eCall*, is the network infrastructure used by the network devices and applications embedded in the cars. In the case of *eCall*, it depends on cell towers, among other facilities, which made researchers investigate other options in the last years. In fact, the investigations began before the proposal of *eCall*, leading to the term Vehicle-to-everything (V2X) [5], a concept related to the ability of vehicles to communicate with pedestrians, other vehicles, the road infrastructure, and the cellular network. V2X communications are expected to be an important part of the Intelligent Transportation Systems (ITS) in the upcoming years enabling the creation of the Internet of Vehicles (IoV). Collision avoidance, traffic control, crash response, and pothole warning are just some of the possible applications of the IoV [6] [7].

In terms of telecommunications, 802.11p [8] and C-V2X [9] are the most established wireless technologies that enable the creation of vehicular networks. However, Long Range (LoRa) radios have also shown good potential due to their low power usage and excellent range [10].

Previous works such as [11] have proven that LoRa modulation is suitable for Vehicular Ad hoc Networks (VANETs), transmitting reliably in distances of up to a few thousand meters, using license-free radio frequencies, and reducing the need for infrastructure. While those authors have stated that LoRa can be useful for safety applications, owing to its previously mentioned characteristics, they have not demonstrated so directly.

In this paper, we propose a crash response system using LoRa-based V2X communications to warn emergency services about a vehicular accident. The system is defined by an event-driven network, broadcast transmissions, and an application protocol. Simulations carried out in the ns-3 simulator [12] suggest that through the use of the proposed system, an

emergency service could be quickly informed of an accident, much faster than it would be by any human being, and without a huge investment in network infrastructure. This could save lives and help to reduce the huge losses caused by road accidents.

As a complementary contribution, all the code produced in this research is publicly available as Open Source Software, allowing others to reproduce the experiments and extend our proposal.

The rest of the paper is organized as follows. Section II presents a brief survey on closely related works. We introduce the system in Section III. Section IV describes the simulation environment used to evaluate the system. Section V describes the results of the performance evaluation. Section VI discusses the findings, and finally Section VII concludes the paper.

II. RELATED WORK

Some previous works in the literature already investigated the usage of LoRa in Vehicular communications [11] [13] [14]. In [11], the authors compared the results obtained from simulating LoRa V2X communications to real experimental data. The metrics obtained from each experiment were Packet Delivery Ratio (PDR), Packet Inter Reception (PIR) time, and Received Signal Strength Indicator (RSSI). The data showed that the RSSI level from the simulations were consistent with the real experiments but it was not possible to reproduce failures caused by interference. We will consider the results obtained in this study when evaluating the performance of our system.

The work in [13] analyzed the resilience of LoRa to the problem of the Doppler shift caused by a fast moving vehicle. Using the results of real experiments and simulations, the authors showed how LoRa parameters should be tuned to reduce symbol time and decrease bit errors. The parameters of our simulation will be similar to the best ones found on this study.

The proposed architecture in [14] is very robust, using Road Side Units (RSUs) to communicate with vehicles in a time-critical and reliable way. However it also shows how costly it can be to limit LoRa routing to single-hop only, since each vehicle has to be in range of a RSU if it wants to communicate with a central server. For the purposes of our event-driven network, simply flooding the network with a broadcast reduces the need for deploying permanent infrastructure.

Table I compares the works summarized in this section with our work. It is possible to observe that our work is the only one focused on developing a new system and using a multi-hop strategy for the delivery of messages.

III. PROPOSAL

Fig. 1 depicts the architecture of our proposed system. The network will be decentralized and comprised of multiple LoRa-enabled devices. There won't be any transmission until an accident happens. For the purposes of this study, we should assume that every vehicle is able to unmistakably detect a crash using onboard sensors. Whenever such an event happens,

TABLE I
COMPARISON TO SIMILAR WORKS.

Ref.	Focus	Scenario	Multi-Hop	Metrics
[11]	Experimental vs Simulation	Vehicular	—	Coverage RSSI PDR PIR
[13]	Performance Evaluation	Vehicular	—	Symbol time Bit errors
[14]	VANET Architecture	Vehicular	—	RSSI PDR Energy
Present work	New System	Vehicular	✓	PDR Coverage Latency

the vehicles involved in the accident will transmit a beacon every few seconds, hoping to reach nearby emergency services.

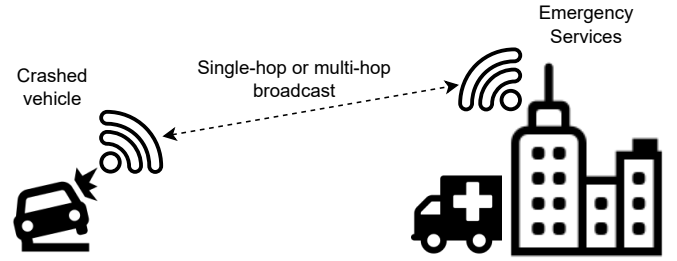


Fig. 1. Architecture of the system

There are three categories of nodes in the network: the vehicle involved in an accident will be named a *caller*, the emergency service which will react to an event will be named a *responder*, and lastly there are the possible *intermediate* nodes between the first two. A *caller* will transmit a *beacon* message every few seconds, until it receives an *acknowledgment* message back.

As stated before, our proposal uses LoRa, which is a modulation technique developed by Semtech, a company based in France, and a founding member of the LoRa Alliance. It uses license-free radio frequencies to enable long range communication for Low Power Wide Area Networks (LPWANs). Through the use of a proprietary chirp spread spectrum modulation, LoRa devices can have ranges of up to 10km, and data rates standing between 0.3kbps and 27kbps [15]. It should be clear that our system was built on top of the LoRa physical layer, and not LoRaWAN¹, which is a much more extensive protocol that implements the data-link and network layers of the OSI model for LoRa communications.

The most important parameter for LoRa modulation is the spreading factor (SF). In a chirp spread spectrum modulation, each symbol, also known as a chirp, is the encoding unit used to transmit data. Lower spreading factors increase the chirp rate, causing faster data rates and shorter range. Conversely, higher spreading factors decrease the chirp rate, meaning slower data rates and longer transmission ranges.

¹<https://loro-alliance.org/about-lorawan/>

The next subsections describe three aspects of our system: the event-driven network, the routing scheme, and the message format.

A. Event-driven Network

The fundamental entity of the network considered in our system is the event. Nodes don't have an identifying address, as messages are not routed to any particular one. Events, on the other hand, have identifying numbers and those are used to route messages across the network.

B. Routing

All messages will be broadcast across the network, avoiding the need for routing protocols and route establishment. Broadcasting is suitable for this scenario, because transmissions are made only when a fairly rare event, a vehicular accident, happens.

Nodes must only be careful to avoid loops, for this reason the protocol must provide the necessary message fields to prevent the retransmission of messages.

C. Message Format

The purpose of the system is to allow the transmission of small messages with the minimal information needed for an emergency response team to arrive at the location of an accident. Considering that, Fig. 2 presents the format of the messages that are going to be transmitted by the vehicles involved in an accident. Each field has the following meaning:

- 1) Event ID (4 bytes): a large integer randomly chosen to identify each event.
- 2) Attempts (2 bytes): this number will be incremented for every message that a node originates. An intermediate node will use this number to avoid loops.
- 3) Hop Count (1 byte): this avoids endless broadcast of a message and should be decremented at every hop.
- 4) Type of Event (1 byte): a node can use this field to inform which type of event it is originating. For example, this could be set to 0 for an accident, 1 for a fire, etc.
- 5) Latitude (4 bytes): float variable that stores the latitudinal coordinate of the event.
- 6) Longitude (4 bytes): float variable that stores the longitudinal coordinate of the event.
- 7) Number of People (1 byte): can be used to help emergency services to estimate the amount of resources that need to be allocated.
- 8) Last known speed (1 byte): can be used by emergency services to estimate the severity of the event.

IV. SIMULATION ENVIRONMENT

In order to validate the implementation and evaluate the performance of the proposed system, the ns-3 network simulator [12] was chosen, since it is one of the most well documented and established network simulators. Another reason for making this choice is its LoRaWAN ns-3 module ², a library

²<https://github.com/signetlabdei/lorawan>

Byte #							
1	2	3	4	5	6	7	8
Event ID				Attempts		Hop Count	Type of Event
Latitude				Longitude			
Number of People	Speed						

Fig. 2. Message fields

that was used to simulate LoRa end-to-end devices communicating directly, without any interaction with gateways.

The propagation loss model *LogDistance* used in [11] proved to be fairly precise in predicting the RSSI between nodes, and so it was used in this work as well. The spreading factor of SF7 was the one used for the simulations, as it has the best data rate and was proven in experimental tests to be the most resilient to Doppler shift caused by moving vehicles [13].

All the code developed during this study is available as Open Source Software on this repository: <https://github.com/batmacumba/loracrashresponse>.

V. RESULTS

The simulator reproduced the main scenario where a *caller* vehicle is involved in an accident, and sends a message across the network to inform a *responder* that it needs immediate medical assistance.

All nodes are static and randomly allocated around a center point, with the maximum distance from the center being 10km to simulate a major urban center. Each test was executed 30 times, and the number of nodes in the network started at 10 and was doubled until it reached 320 nodes. For each execution, the position of the nodes was shuffled so that a new network topology was simulated. Fig. 3 presents the distribution of the nodes in a generated network instance with 100 nodes.

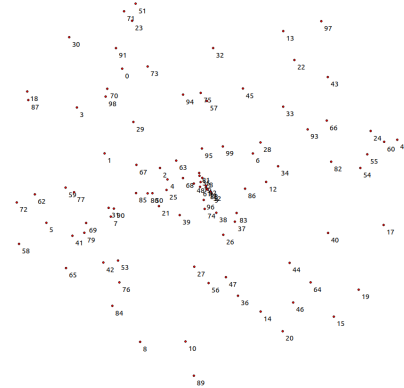


Fig. 3. Sample network – 100 nodes

The PDR, Latency, and Network Coverage presented in the next subsections are the average of the values obtained in all test instances. The *caller* node originated a single beacon, without any further attempts, so that the test results would be more revealing.

A. Packet Delivery Ratio

This test was intended to measure the number of times that the *responder* actually received the message originated by the *caller*. To calculate the PDR, the formula was:

$$\text{PDR} = \frac{\text{number of received beacons}}{\text{number of beacons sent}}$$

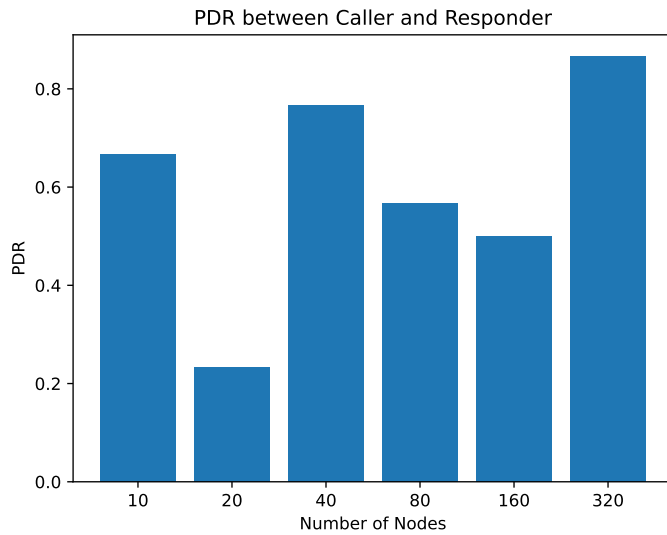


Fig. 4. Packet Delivery Ratio

It is possible to see in Fig. 4 that there is a slight positive correlation between the number of nodes and the PDR in this scenario. The density of the network is the most probable factor here: with fewer nodes, the distance between them tends to be bigger, and this in turn causes lower RSSI and more packet loss. With only 10 nodes distributed around the test area, the originating node sometimes was isolated and could not transmit to any other node.

As the number of nodes increased, so did the PDR, although there were ratios lower than expected at $N=20$, $N=80$, and $N=160$.

On a real life scenario, the *caller* would retransmit the beacon as many times as necessary before receiving an acknowledgment, which should dramatically improve the probability of it being delivered to the *responder* node.

B. Latency

Another interesting result is the latency measured until the *responder* node received the message (Fig. 5). This shows whether the network is able to handle low latency applications such as this one.

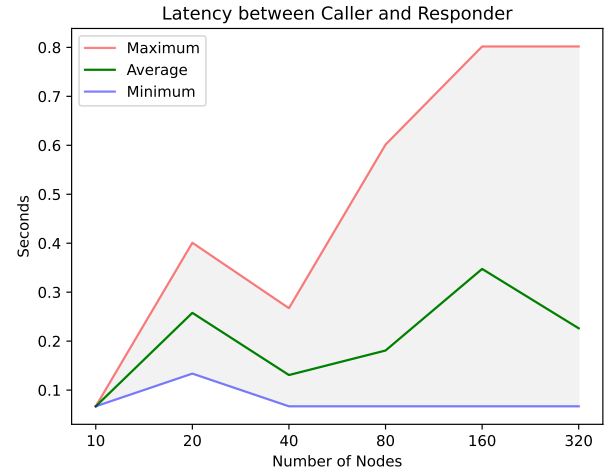


Fig. 5. Latency

The average latency increased as the network density increased. This is an important observation, and could be related to interference between broadcasting nodes. Since pure LoRa is an ALOHA protocol, nodes could be trying to transmit at the same time, and this could be causing increased latency.

C. Network Coverage

To have a better understanding of what's happening across the whole network, not only with the *caller* and *responder* nodes, one can look at the graph in Fig. 6. To calculate the coverage, the formula was:

$$\text{Network Coverage} = \frac{\text{number of nodes that received the beacon}}{\text{total number of nodes}}$$

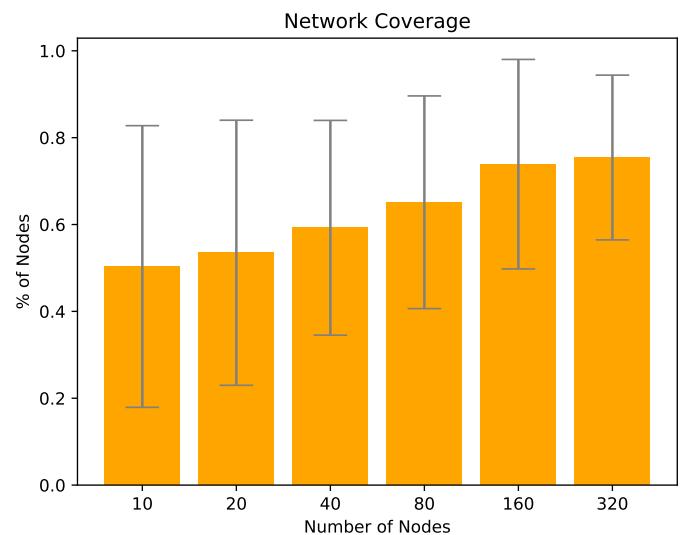


Fig. 6. Network Coverage

It shows that network density is a decisive factor in the ability of this network to propagate a broadcast across all nodes.

VI. DISCUSSION

The above results are very positive as they show that LoRa V2X communication is effective in transmitting messages using multi-hop routes travelling over several kilometers, without inducing too much latency.

A very valid question raised by the results is the one of the medium access protocol of pure LoRa. It looks as if interference could be a major issue if this network was scaled to a large number of nodes. More work could be done in this context, either by measuring the packet loss due to interference, or by greatly increasing the number of nodes in the simulation. Carrier-sensing techniques could be analyzed as possible solutions to mitigate this problem.

The results should also be interpreted with caution, as the simulation did not take into account the heavy interference that would be caused by an urban environment, and the Doppler shift caused by the mobility of nodes. For further work, it would be advisable to improve the propagation loss model to include more interference, and to simulate using moving nodes.

A sensitive matter that should also be dealt with in such an application is the confidentiality and authenticity of the information transmitted in the messages. A real-world application would have to apply cryptography techniques to ensure that private individuals could not have access to the geographical position of an accident and use this information to their advantage.

Additionally, emergency services would also have to be able to check the authenticity of received beacons, in order to avoid being deployed to fake calls, and to prevent the modification of beacon messages by intermediate nodes. Ideally, every node should be able to verify the authenticity of the beacons, so that unauthorized transmissions would not be retransmitted.

VII. CONCLUSION

This study has successfully shown that a crash response application based on LoRa V2X messages can be implemented to quickly inform emergency services of a vehicular accident. The latency of this application is very low, and is able to significantly reduce the amount of time it takes for medical care to arrive at the scene of an accident.

Considering that LoRa has very good range for vehicular applications, and that this application has very low data rate requirements, this result is not unexpected. Previous works have already shown that LoRa is suitable for VANETs, although most of the existing literature focus on single-hop communications.

A major benefit of the proposed architecture is the exemption from deploying previous infrastructure on the roads. This is guaranteed in densely populated areas but could be a problem in non-urban areas, where vehicles could be distant from each other, beyond the range of LoRa.

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